FitNet

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Introduction

Poor exercise form is a widespread issue in modern fitness, especially among beginners who lack access to proper coaching. Incorrect technique often leads to the development of bad habits that are difficult to unlearn over time, reducing workout effectiveness and increasing the risk of injury. In fact, approximately 460,000 exercise-related injuries were reported last year, many of which were linked to improper form and preventable mistakes. These injuries not only limit physical progress but can also discourage individuals from continuing a consistent fitness routine.

To address this gap, this project proposes a more accessible and practical solution: a mobile application that uses computer vision and machine learning to analyze exercise form directly from user-recorded video. By leveraging pose estimation and angle-based motion analysis, the system can provide immediate, personalized feedback without the need for costly hardware or paywalled services. This approach aims to improve workout safety and make high-quality form guidance available to a broader population.

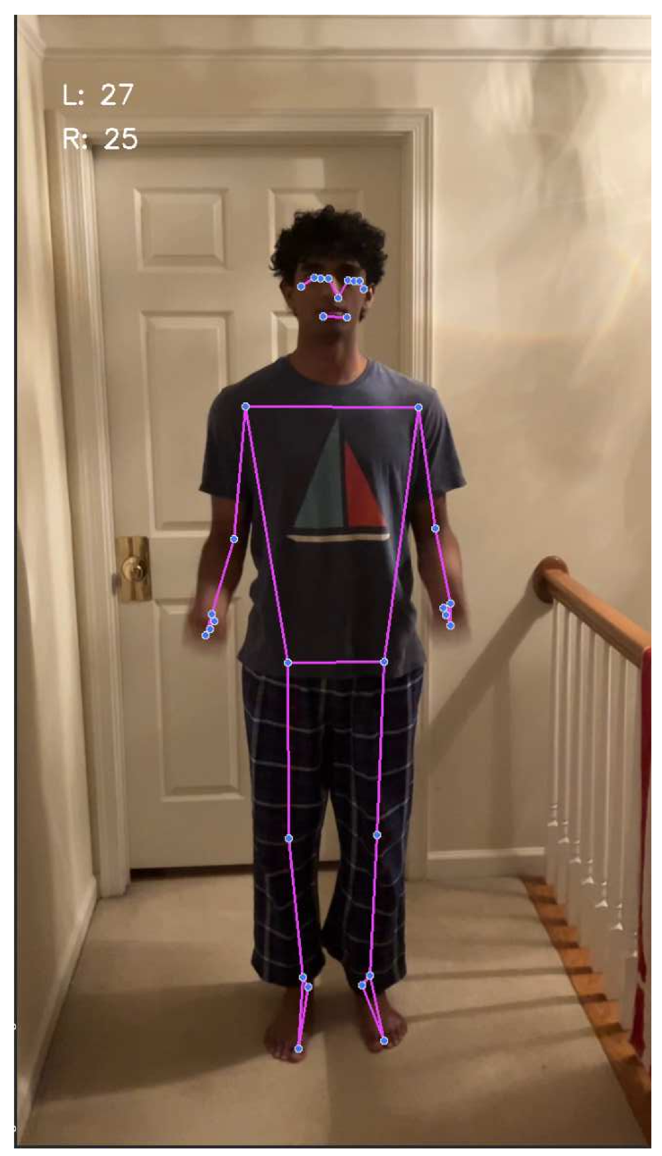
Methods

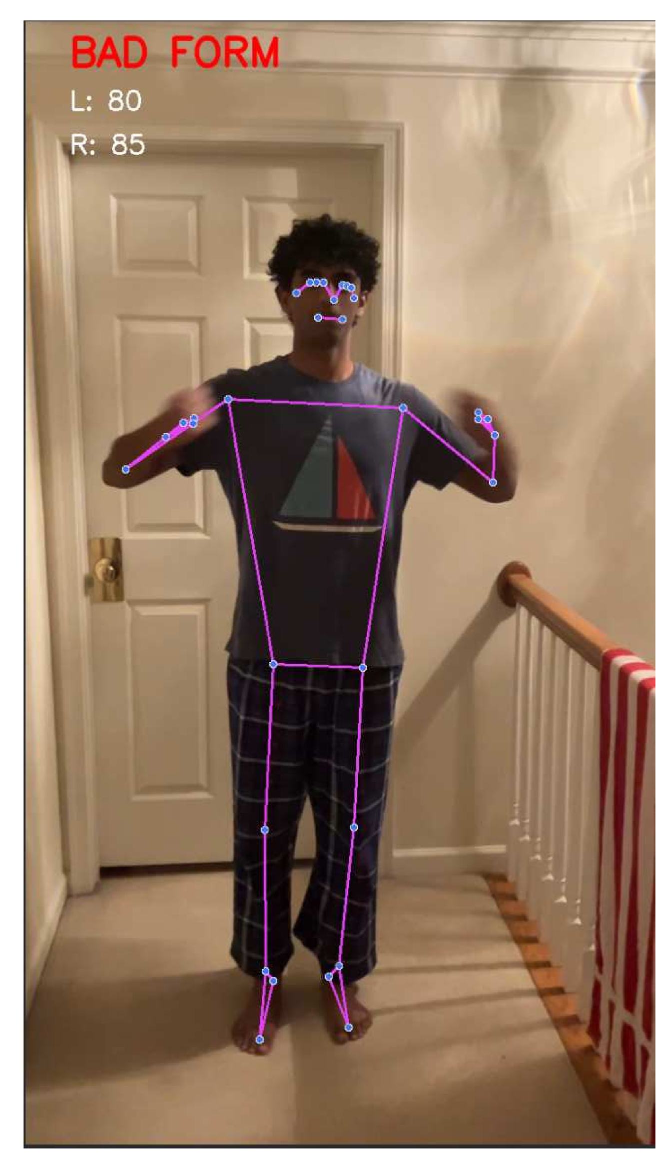
This project used an iterative development process that involved MediaPipe pose estimation, angle-based motion analysis, and mobile app user interface design. The system's function was to evaluate exercise form in real time and provide corrective feedback to the user. The exercise to be used was first targeted towards bicep curls, as the movement is a repetitive, simple motion around one joint. This rendered it ideal for system calibration and verification of correct angle detection before progressing to more complex exercises.

MediaPipe was used to recognize key body landmarks from bicep curl videos at the shoulder, elbow, and wrist. The program calculated the elbow joint angle by employing trigonometric equations based on the coordinates of these landmarks. Measured angles for every frame were contrasted with a reference range for proper form. When the angle exceeded a specified limit, the system identified it as defective and provided feedback such as "place your elbow closer to your side."

Findings

The prototype application successfully analyzed bicep curl form using real-time pose tracking and angle measurement. The system was able to detect key arm joints (shoulder and elbow) and calculate shoulder angles during each repetition. Based on these measurements, the app correctly identified whether a repetition followed acceptable form or showed clear errors, such as excessive body movement or inconsistent elbow positioning.

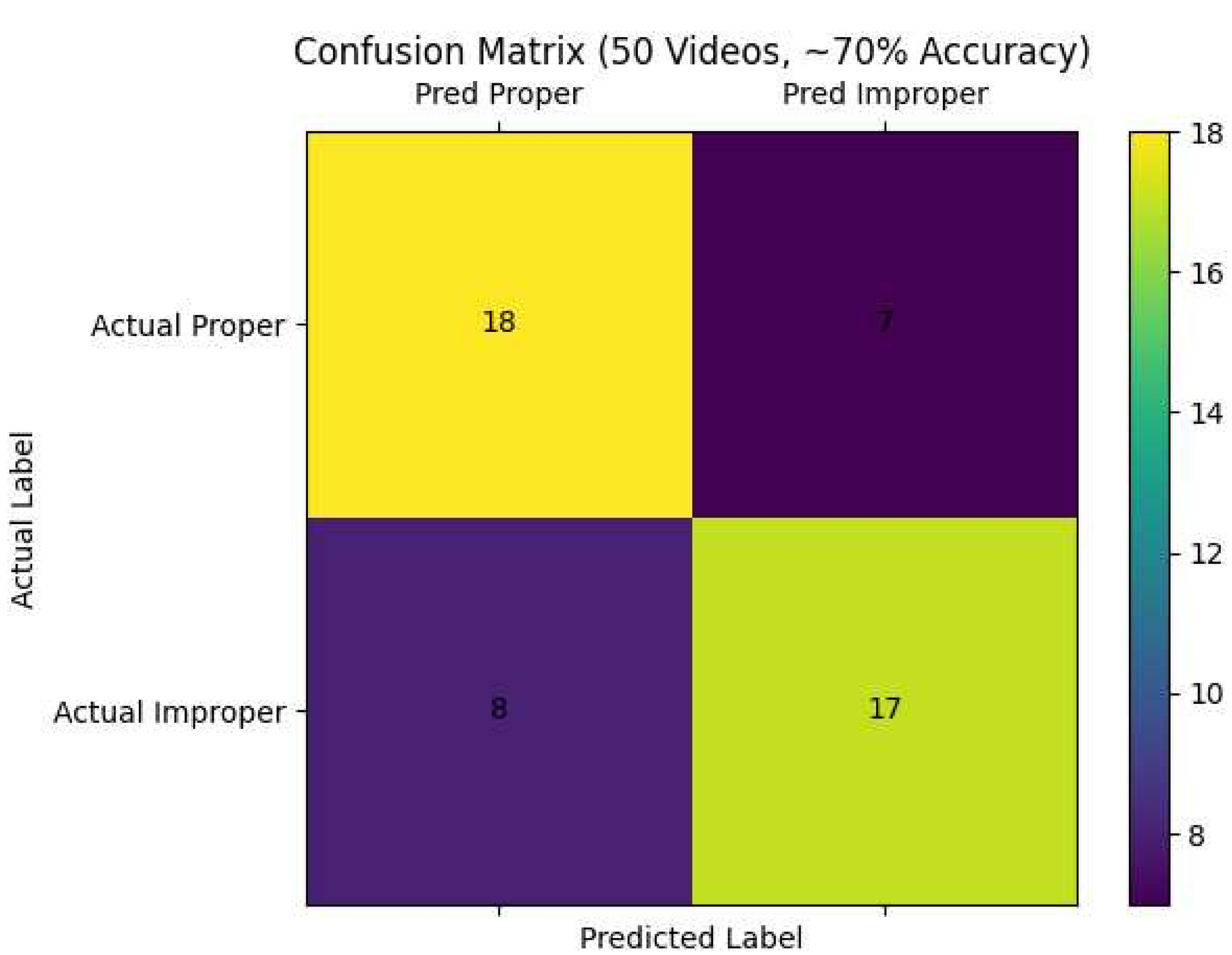




Testing was conducted using recorded videos that included both correct and incorrect bicep curls. The system consistently identified obvious form issues when the elbow angle moved outside the expected range or when motion patterns suggested momentum rather than controlled lifting. In controlled settings with a clear camera view, the app performed reliably and produced feedback at appropriate times during the exercise.

Example Classification

The confusion matrix shows that the model achieved 70% accuracy, correctly classifying 35 out of 50 videos. It performed similarly on both proper and improper form, with 18 and 17 correct predictions respectively. However, some misclassifications still occurred, including 7 false negatives and 8 false positives. False negatives are more concerning because they allow incorrect form to go uncorrected. Overall, the results show moderate reliability, but improvements are needed to reduce errors and improve real-world performance.



Future Research

To address this limitation, future work will focus on body segmentation to estimate user body type more accurately. By adjusting angle thresholds based on segmented body proportions, the system can better personalize feedback and reduce false classifications. This improvement is important because the optimal bicep curl angle naturally varies from person to person.

Additional future work includes expanding the system to support more exercises, testing with a larger group of users, and improving the user interface. Further research questions include how personalized angle ranges affect injury prevention and whether adaptive feedback improves long-term exercise habits. Overall, this project demonstrates that lightweight computer vision techniques can meaningfully enhance fitness guidance in the real world.

Works Cited

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